

Network Security

Q) Explain Secure Socket Layer (SSL)?

SSL stands for **Secure Socket Layer**. It is a **security protocol** that helps to create a **secure communication** channel between a **web browser (client)** and a **web server** over the internet.

It protects data using **encryption**, which means it converts the information into secret code so that hackers can't read it.

It is mostly used in:

- Online shopping websites (e-commerce)
- Internet banking
- Login pages
- Email services

Main Goals of SSL :-

1. **Encryption** – Protects data from hackers during transmission
2. **Authentication** – Confirms that the server is real (not fake)
3. **Data Integrity** – Ensures data is not changed during transfer

How SSL Works (with Steps) :-

1. Client Hello:

- Browser sends a request to the server to start a secure session.
- It sends SSL version, supported ciphers, and a random number.

2. Server Hello:

- Server responds with its own random number and SSL certificate (includes public key).

3. Certificate Verification:

- Browser checks if certificate is valid (signed by a trusted Certificate Authority).

4. Session Key Creation:

- Browser creates a session key and encrypts it using the server's public key.

5. Secure Communication Begins:

- Both browser and server now use this session key to encrypt and decrypt the data.

SSL Protocols (Layers in SSL) :-

Handshake Protocol	Change Cipher Spec Protocol	Alert Protocol	HTTP
SSL Record Protocol			
TCP			
IP			

1.Handshake Protocol:

- This protocol helps to **start a secure session**.
- It involves exchanging keys, verifying server identity, and creating a shared session key.

2. Record Protocol:

- This protocol is used to **send and receive actual data securely**.
- It uses encryption (like AES) and adds MAC (Message Authentication Code) to keep data safe.

3. Alert Protocol:

- If there's any problem or error, this protocol sends **alert messages** (like handshake failure or certificate error).

4. Change Cipher Spec Protocol:

- It informs both parties that future messages will be encrypted with the **new session key**.

Advantages of SSL:

- Ensures **secure online transactions**
- Builds **trust** with users
- Protects against **phishing and data theft**
- **Mandatory** for all modern websites (shows  padlock)

Q) Explain Active and Passive Attack with Examples.

In Network Security, an **attack** means an attempt by an unauthorized person (called an **attacker** or **hacker**) to access, damage, or steal information from a system or network.

There are two main types of attacks :----

1. Active Attack :-

An **active attack** is when an attacker changes, damages, or interferes with the data during transmission.

The attacker does not just read, but modifies or creates false data.

Goal of Active Attack:

- To **harm the system**, change the data, or **disrupt communication**.

Examples:

1. **Modification of Messages** – Changing the content of a message.
2. **Denial of Service (DoS)** – Overloading the system to make it crash.
3. **Man-in-the-Middle (MITM)** – Attacker places himself between two parties to modify or steal data.
4. **Spoofing** – Attacker pretends to be someone else (like fake IP address or fake email).

Result:

- It **affects data integrity**.
- It can **crash systems or steal control**.

2. Passive Attack :-

A **passive attack** is when an attacker **only observes or listens** to the communication between two systems, but **does not change** anything.

Goal of Passive Attack:

- To **collect secret information** like passwords, messages, or private keys.

Examples:

1. **Eavesdropping** – Listening to private communication between users.
2. **Traffic Analysis** – Studying the data flow (like size, timing) to get useful information.
3. **Monitoring Emails or Files** silently.

Result:

- It **affects confidentiality**.
- Hard to detect because **nothing is changed**.

Difference Between Active and Passive Attacks :-

Feature	Active Attack	Passive Attack
Action	Data is modified or affected	Data is only observed or captured
Detection	Easy to detect	Hard to detect
Goal	To harm or damage the system	To steal information silently
Example	DoS, Spoofing, MITM	Eavesdropping, Traffic Analysis
Security Impact	Integrity & Availability	Confidentiality

Q) How do Security Mechanisms protect a system?

In Network Security, security mechanisms are special tools, methods, or techniques used to protect computer systems, networks, and data from attacks, threats, and unauthorized access.

They help in maintaining:

- **Confidentiality** (keeping data secret)
- **Integrity** (keeping data unaltered)
- **Availability** (keeping system working)
- **Authentication** (confirming identity)
- **Non-repudiation** (no denial of actions)

Types of Security Mechanisms and How They Protect :-

1. Encryption:

- Converts plain text into unreadable form (**ciphertext**) using a key.
- Even if an attacker gets the data, they cannot read it.
- Protects **confidentiality** of data.

Example:

HTTPS websites use SSL/TLS encryption to protect user data like passwords and card details.

2. Authentication:

- Verifies the identity of a user, system, or device.
- Only authorized users can access the system.
- Uses passwords, biometrics, OTP, ID cards, etc.

Example:

Logging in to Gmail with a password + OTP on phone.

3. Access Control:

- Controls who can access what resources.
- Defines permissions like read, write, delete.
- Prevents unauthorized access or misuse.

Example:

In an office system, employees can only access their own department's files.

4. Digital Signatures:

- Adds a unique digital code to a document or message.
- Proves the **sender's identity** and ensures data is not changed.
- Used in legal documents, emails, or software.

Example:

PDF files with digital signatures from official authorities.

5. Firewalls:

- A software or hardware that filters incoming and outgoing network traffic.
- Blocks unauthorized access from the internet.
- Acts like a security guard for your system.

Example:

A firewall blocks hackers from entering your home Wi-Fi network.

6. Intrusion Detection Systems (IDS):

- Monitors system or network for suspicious activities.
- Alerts the admin if it detects something wrong.
- Helps to catch unauthorized or harmful actions quickly.

7. Antivirus and Anti-malware Software:

- Detects and removes viruses, malware, ransomware, spyware, etc.
- Keeps the system safe from harmful software.
- Provides real-time protection.

8. Audit Logs:

- Keeps a record of all user activities in a system.
- Helps to detect if any unauthorized or illegal activity occurred.
- Useful during investigations or system monitoring.

Q) Write notes on :-

1) Digital Signature

2) Firewall

1> What is a Digital Signature :----

A Digital Signature is like an **electronic fingerprint** used to verify the identity of the sender and ensure that the message or document has not been changed.

It provides authentication, data integrity, and non-repudiation in digital communication.

It is used in :

- Emails
- PDF documents
- Software updates
- Online contracts

Why is a Digital Signature Important?

- It proves that the message really came from the sender.
- It ensures that the message is not changed during transmission.
- It prevents the sender from denying sending the message (non-repudiation).

How Digital Signature Works (Simple Steps):

1. Message Creation:

The sender writes the message or document.

2. Hash Function is Applied:

A hash value (unique fixed-size output) is generated from the message.

3. Encrypting the Hash:

The sender encrypts the hash using their private key.

This encrypted hash is the digital signature.

4. Sending:

The original message + digital signature is sent to the receiver.

5. Verification:

The receiver uses the sender's public key to decrypt the signature.

Then generates the hash of the received message again.

If both hashes match → Signature is valid.

Conceptual Example -----

Suppose Ravi sends a signed contract to Rahul:

- Ravi's software generates a **digital signature** from the contract using Ravi's **private key**.
- Rahul receives it and verifies it using Ravi's **public key**.
- If the verification passes, Rahul knows:
 - The message is from Ravi (authentication).
 - The message is not changed (integrity).
 - Ravi can't deny sending it (non-repudiation).

Features of Digital Signature:

Feature	Description
Authentication	Proves sender's identity
Integrity	Ensures message has not been changed
Non-repudiation	Sender cannot deny the message later
Secure	Uses cryptography for protection

2> What is a Firewall :-----

A **firewall** is a **security system** (hardware or software) that **monitors and controls** the incoming and outgoing traffic on a network based on **predefined security rules**.

Its main job is to **protect your system or network** from **unauthorized access, hackers, and malware**.

Why Firewall is Important ?

- It acts like a **security guard** for your computer/network.
- It checks every data packet that enters or leaves the network.
- If the data looks dangerous or unauthorized, the firewall **blocks it**.

Types of Firewalls :-

1. Packet Filtering Firewall:

- Checks packets based on IP address, port number, and protocol.
- If the packet does not meet the rule → it's blocked.
- Fast but basic level of security.

2. Stateful Inspection Firewall:

- Tracks **state of connection** (not just individual packets).
- More secure than packet filtering.

3. Proxy Firewall:

- Works as a **middleman** between user and internet.
- Hides the actual network from outsiders.
- Slower but very secure.

4. Next-Generation Firewall (NGFW):

- Advanced firewall with features like:
 - **Deep packet inspection**
 - **Application filtering**
 - **Intrusion prevention**
 - **Malware detection**

5. Software Firewall:

- Installed on a single computer (example: Windows Defender Firewall).
- Protects that specific device.

6. Hardware Firewall:

- A physical device used to protect a whole network.
- Common in businesses or large systems.

Functions of a Firewall:

- **Monitors Traffic:** Allows safe data and blocks harmful data.
- **Prevents Unauthorized Access:** Stops hackers and suspicious users.
- **Protects Against Viruses and Malware:** Can block known threats.
- **Controls Internet Use:** Admin can block websites or limit traffic.
- **Logs Activities:** Keeps records of blocked attempts and security events.

Advantages of Firewall:

- Improves network security
- Protects against cyber attacks
- Helps in monitoring and control
- Prevents data theft or misuse

Q) Explain Intruders and Viruses ?

What are Intruders :::::-----

An **intruder** is a person or a program that **tries to break into a computer system or network** without permission.

The main goal of an intruder is to:

- Steal data
- Damage systems
- Change or delete information
- Install malware secretly

Types of Intruders -:

1. ♦ Masquerader:

- This is an **outsider** (not an authorized user).
- He tries to enter the system by **pretending to be someone else**.
- For example, a hacker who guesses or steals someone's password and logs in.

2. ♦ Misfeasor:

- This is an **insider** (like an employee or user who already has access).
- He uses his own access to **do harmful things**, such as copying confidential files or deleting important data.
- Example: An employee leaking company information.

3. ♦ Clandestine User:

- This intruder is someone who tries to **hide his presence** in the system.
- He **secretly controls system functions** without the admin knowing.
- Example: A person who disables security logs and runs illegal programs.

How to Prevent Intruders:

- Use strong passwords
- Enable multi-factor authentication
- Install firewalls and Intrusion Detection Systems (IDS)
- Keep software updated
- Monitor logs regularly

What is a Virus :::::-----

A virus is a type of malicious software (malware) that attaches itself **to a** file or program and spreads from one system to another, often without the user's knowledge.

It usually attaches itself to files, and when the file runs, the virus activates and starts causing damage.

How Viruses Work:

- A virus is usually hidden inside software, email attachments, or downloads.
- When the infected file is opened, the virus gets activated.
- It may then damage files, slow down the system, or spread to other computers.

Types of Viruses -:

1. ♦ File Infector Virus:

- This virus attaches to executable files (.exe).
- When the infected file is opened, the virus runs and spreads to other programs.
- Example: A virus that hides in a software setup file.

2. ♦ Macro Virus:

- It targets files created by Microsoft Word, Excel, etc.
- These files use “macros,” and the virus hides in those.
- When you open the document, the virus activates.
- Example: A Word file received in email that corrupts your system.

3. ♦ Boot Sector Virus:

- It infects the boot sector of a storage device (like hard disk or pen drive).
- It starts working as soon as the system boots up.
- These viruses are very dangerous and can make your system unbootable.

4. ♦ Worm:

- A worm is like a virus but it spreads on its own without attaching to files.
- It can copy itself across networks and use up system resources.
- Example: A worm that sends itself through emails to everyone in your contact list.

5. ♦ Polymorphic Virus:

- This virus **changes its code** every time it spreads.
- It's very hard to detect because antivirus software can't easily recognize it.
- It's used to attack large systems secretly.

Effects of Viruses:

- Slow system performance
- Delete or corrupt files
- Crash the operating system
- Steal personal or financial information

Protection from Viruses:

- Install a good antivirus and keep it updated
- Never open unknown email attachments or links
- Regularly back up important data
- Don't download pirated software

Q) What do you mean by malware? Explain in details.

Malware stands for Malicious Software. It is any software or program that is designed to harm, damage, steal, or control a computer system or network without the user's permission.

The main purpose of malware is to:

- Steal information
- Damage systems
- Spy on user activity
- Control devices remotely

Malware is created by hackers or cyber criminals to attack individuals, companies, or even government systems.

How does Malware Spread :-

- Through email attachments
- From infected websites
- By downloading pirated software or games
- Through USB drives or external devices
- From social media links or pop-up ads

Types of Malware :::: ----

♦ **1. Virus :**

A virus is a program that attaches itself to files or programs and spreads when the infected file is opened.

It can delete files, corrupt data, and slow down your system.

Example:

A virus hidden in a game setup file.

♦ **2. Worm :**

A worm spreads by itself, without attaching to files.

It uses **network connections** to copy itself to other computers and causes network traffic and slowdowns.

Example:

A worm that spreads via email to every contact in your list.

◆ 3. Trojan Horse :

A Trojan looks like a normal program but it secretly does harmful activities like giving control of your PC to a hacker.

It does not replicate like a virus or worm.

Example:

A free game or app that secretly steals your passwords.

◆ 4. Spyware :

This malware spies on your activity and collects your personal data like keystrokes, passwords, and browsing habits.

It works silently in the background.

Example:

A tool that records everything you type (keylogger).

◆ 5. Adware :

Adware shows unwanted ads on your screen, often pop-ups or redirects.

It may not always be harmful but can be annoying and sometimes bring more malware.

Example:

Browser getting redirected to random shopping or adult sites.

◆ 6. Ransomware :

Ransomware locks your files or system and demands money (ransom) to unlock them.

It is very dangerous and often used in cybercrime.

Example:

WannaCry ransomware attack in 2017 that locked hospital files across many countries.

◆ 7. Rootkits :

Rootkits give remote control of your computer to attackers without your knowledge.

They hide themselves deeply inside your system and are hard to detect.

Example:

A hacker controlling your webcam or deleting system files without you knowing.

How to Protect Against Malware :-

- Install and update antivirus or anti-malware software
- Don't open unknown email links or attachments
- Avoid pirated software or cracks
- Keep your system and apps updated
- Don't connect untrusted USB drives
- Use strong passwords and two-factor authentication